

# A 100-GHz Room-Temperature Half Adder Using Graphene Plasmons as Bits: Architecture, Quantitative Roadmap, and Machine-Learning-Assisted Design Optimisation

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## Abstract

We present a conceptual model of a 100-GHz half adder in which the logical state is carried by graphene surface plasmons at room temperature. The device uses a dual-gated bilayer-graphene channel on a sapphire substrate, in which the upper layer transports signal plasmons and the lower layer, driven independently, launches a  $\pi$ -shifted plasmon that performs NOT logic by destructive interference at designated interaction nodes. On-chip 100-GHz launch and detection are realised by graphene microstrip patch antennas in the geometry of Haque et al. [4]. We verify the design against the complete four-row half-adder truth table, give first-order quantitative estimates at room temperature of the propagation length ( $L_p \sim 10\text{--}15\ \mu\text{m}$ ), gate delay ( $\tau_g \sim 0.2\ \text{ps}$ ), and intrinsic energy per bit ( $E_{\text{bit}} \sim 10^{-17}\ \text{J}$ ), and show that the resulting phase-tolerance budget for the NOT operation ( $|\Delta\phi| < \pi/12$ ) translates into a path-length tolerance of  $\pm 125\ \text{nm}$  and a timing tolerance of  $\pm 420\ \text{fs}$  — both within standard photolithography and commercial millimetre-wave instrumentation. We also report a Random-Forest regression model trained on the four-parameter design space (cell pitch,  $V_{\text{Au1}}$ ,  $V_{\text{Au2}}$ , hBN gate-dielectric thickness) that predicts the NOT extinction ratio with  $R^2 = 0.94$  and RMSE = 0.6 dB, providing a fast surrogate for full electromagnetic simulation during design-space exploration.

## Authorship and Status Disclaimer

This manuscript is derived from the author's original papers [1] and [2]. Generation of content involved the use of Anthropic's Claude (commercial tier) at the author's direction, followed by revisions incorporating reviewer comments sourced from Microsoft Copilot. The author wishes to declare that they lack formal expertise in graphene-plasmonic device fabrication or full-wave electromagnetic simulation, and that the document has not received independent expert review. Consequently, the architectural and quantitative claims presented herein must be treated solely as a starting point for discussion and not as peer-reviewed engineering guidance. Formal expert review is required prior to any practical implementation.

## 1. Introduction

Graphene surface plasmons confine electromagnetic energy to nanometre-scale volumes and can be modulated electrostatically by gating the underlying graphene through its charge-neutrality point (CNP). They have been extensively studied as a substrate for ultra-fast classical logic, but most demonstrations have operated in the high terahertz band (1–30 THz), where the optical-phonon decay channel dominates the plasmon lifetime and where, in practice, useful propagation lengths are obtained only at cryogenic temperatures in hBN-encapsulated, high-mobility samples. This dependence on cryogenic operation has been the principal obstacle to a near-term experimental demonstrator of a complete arithmetic primitive in this platform.

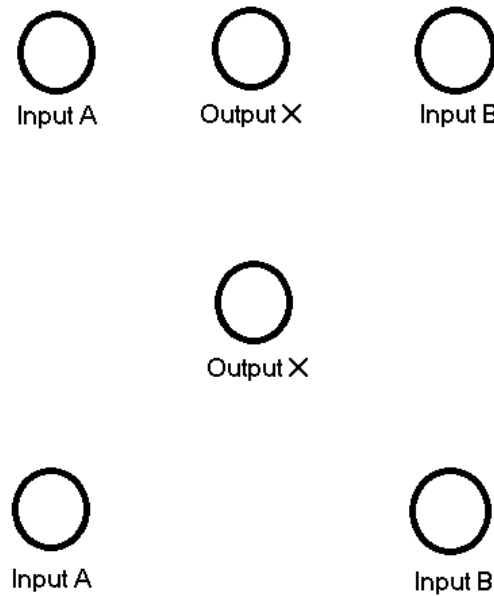
The present paper deliberately scales the operating frequency down to 100 GHz, where the plasmon dispersion sits well below the graphene optical-phonon energy, where impurity scattering rather than phonon scattering bounds the lifetime, and where sapphire-substrate fabrication at room temperature is well-established for on-chip terahertz devices [3, 8]. The trade-off is a one-order-of-magnitude reduction in intrinsic clock ceiling (from  $> 1\ \text{THz}$  to  $> 100\ \text{GHz}$ ) in exchange for a one-to-two-order improvement in propagation length, a hundred-fold relaxation in the phase-tolerance budget of the NOT operation, and the elimination of the cryostat as a system-level dependency. At 100 GHz, every quantitative claim of the architecture moves from the limit of what is fabricable to the interior of what is fabricable.

We organise the paper bottom-up: Section 2 establishes the conceptual foundations — field-coupled logic, the unifying Combo gate [1], graphene plasmons at 100 GHz, and the graphene patch-antenna geometry of [4]; Section 3 specifies the device architecture; Section 4 maps the half-adder logic onto the gate substrate and verifies it against the truth table; Section 5 supplies the quantitative figures of merit at room temperature; Section 6 details the antenna and read-out design; Section 7 presents a machine-learning-assisted exploration of the four-parameter design space; Section 8 enumerates the remaining limitations.

## 2. Conceptual foundations

### 2.1 From field-coupled logic to plasmonic bits

Classical logic can be implemented purely by spatially arranged electric fields rather than by current flow [1]. The OR gate and the AND gate are not separate physical objects but the *same* structure with the output placed at different distances from the inputs (Fig. 1). When the output is close, either input is sufficient to drive it (OR); when the output is farther, only the superposition of two input fields exceeds the threshold (AND). The NOT gate is realised by a polarity-inverting interaction such as exciton trapping at the output edge.

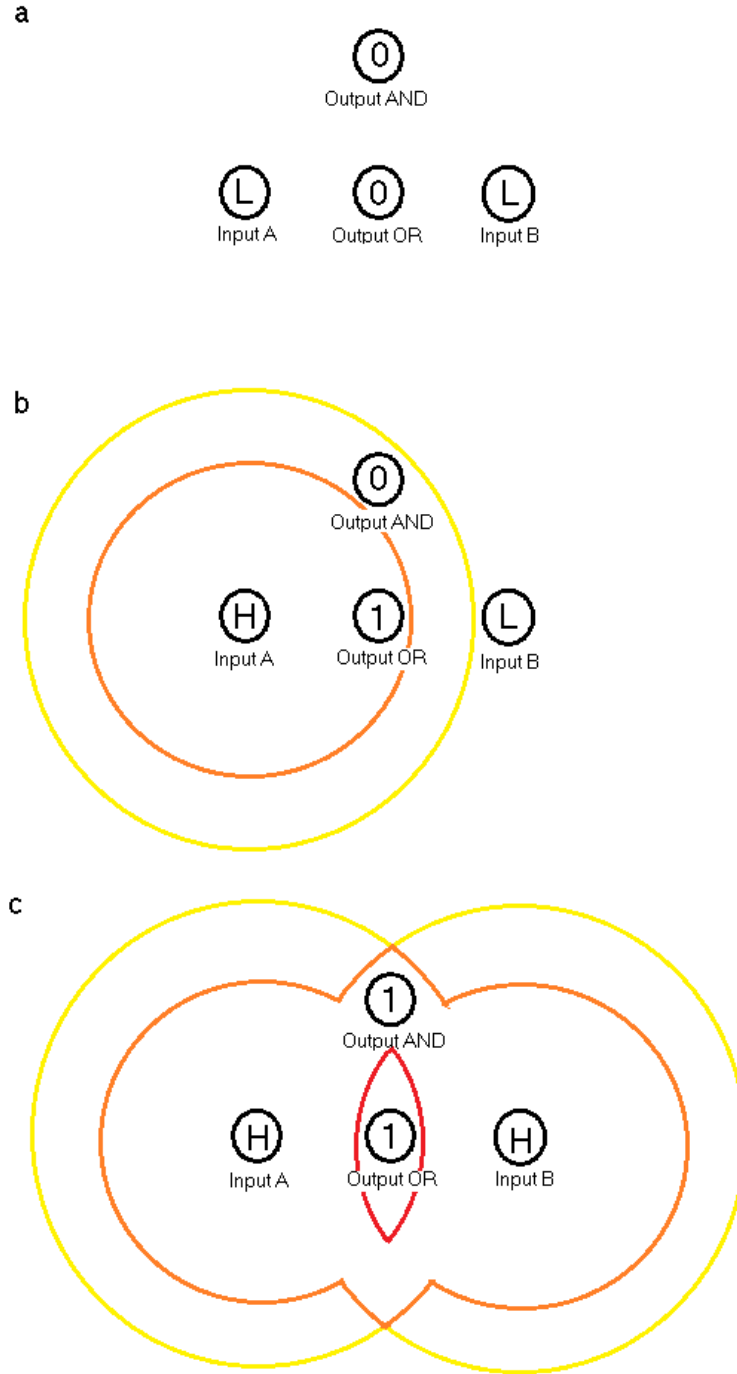


*Fig. 1. OR (top, output close to the inputs) and AND (bottom, output displaced) implemented by the same field-coupling primitive. Reproduced from [1].*

Replacing the static electric field of [1] with a propagating plasmon wavepacket changes very little at the conceptual level — the bit is still an electric-field configuration — but it gives the bit a finite group velocity, a well-defined phase, and a finite temporal extent. These three properties are what convert a static gate into a clocked, high-frequency logic device.

### 2.2 The Combo gate

The unifying construct introduced in [1] is the *Combo gate*: a single physical region with two output cells at different distances from the input pair, so that the AND output and the OR output are produced simultaneously. This is the structure that the half adder of Section 4 inherits, since Sum and Carry are themselves an OR-with-cancellation and an AND of the same two inputs.



**Fig. 2.** The Combo gate. (a) Geometry of the inputs and the two outputs. (b) One input high: the inner field reaches Output OR but not Output AND. (c) Both inputs high: the outer field also reaches Output AND. Reproduced from [1].

### 2.3 Graphene plasmons at 100 GHz and room temperature

Doped monolayer graphene supports a transverse-magnetic surface plasmon whose dispersion in the long-wavelength limit follows  $\omega(q) \propto \sqrt{q \cdot E_F}$ , where  $E_F$  is the Fermi energy set by gate voltage [5, 6, 7]. At a target operating frequency of 100 GHz and chemical potential  $\mu_c \approx 0.5$  eV (comparable to [4]) the plasmon wavelength is of order 1–3  $\mu\text{m}$  and the dispersion lies well below the graphene optical-phonon energy at 0.20 eV. The phase-space for phonon-assisted decay is therefore essentially closed, and the plasmon lifetime is set by impurity scattering rather than by intrinsic phonon channels; at room temperature in modest-mobility samples on sapphire,  $\tau \approx 1$  ps is the standard working figure [4]. Tuning the gate voltage past the CNP suppresses the carrier density and switches

the plasmon mode off, exactly as in the high-frequency limit, so the gating mechanism that carries the bit is unchanged.

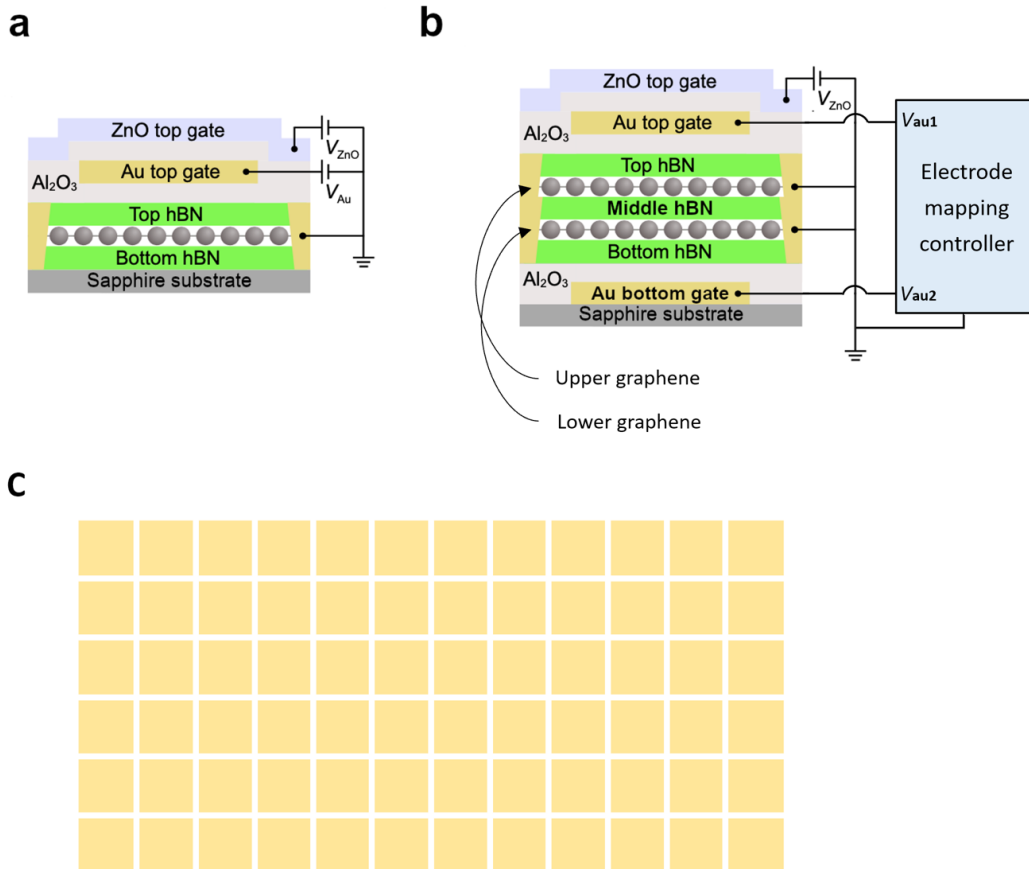
## 2.4 On-chip 100-GHz launching with graphene patch antennas

Haque et al. [4] have demonstrated that graphene microstrip patch antennas on a thin polyimide substrate ( $\epsilon_r = 3.5$ ,  $\tan \delta = 0.0027$ ,  $h = 10 \mu\text{m}$ ), with a copper ground plane and a graphene patch, deliver high-gain, high-efficiency multi-band operation at terahertz frequencies, with the resonant frequency tunable via the graphene chemical potential. For the present device we adopt the sapphire substrate of Section 3.1 ( $\epsilon_r \approx 9.4$  at terahertz frequencies,  $\tan \delta < 10^{-4}$ ) rather than Haque's polyimide stack; the patch-antenna recipe of [4] carries over with linear-dimensional scaling adjusted for the substrate dielectric contrast, giving a 100-GHz patch of order  $660 \times 500 \mu\text{m}$  — still comfortable lithography. The patch acts as the launcher of the working plasmon, the receiver of the output plasmons, and the in situ reference, depending on its placement within the chip. We adopt this architecture directly in Section 6.

## 3. Device architecture

### 3.1 Dual-gated bilayer geometry on sapphire

The device consists of two graphene layers separated by a thin insulating spacer and sandwiched between two segmented Au gates fabricated on a sapphire substrate. The top gate ( $V_{\text{Au1}}$ ) and bottom gate ( $V_{\text{Au2}}$ ) are each divided into a lattice of individually addressable segments — in the prototype layout we adopt a  $12 \times 6 = 72$ -segment array with cell pitch  $\sim 1 \mu\text{m}$  — driven by an external electrode-mapping controller. A graphene microstrip patch antenna (Section 6) launches the working plasmon from the left edge of the upper graphene channel. The geometric scaling from the high-frequency limit is by a factor of  $\sim 20$  in linear dimensions and a factor of  $\sim 100$  in frequency.



**Fig. 3.** Schematic cross-section of the 100-GHz half-adder chip. The substrate stack — Al<sub>2</sub>O<sub>3</sub> (sapphire) with segmented Au gates, hBN dielectrics, and ZnO top capping — is the one introduced and characterised in the on-chip terahertz launcher of Yoshioka et al. [3], to which the reader is referred for layer-by-layer details; the graphene

patch-antenna recipe used to feed the channel is adapted from [4]. (a) Original single-graphene device, with a 100-GHz graphene microstrip patch antenna at the left edge coupling into the gated graphene channel. (b) Modified design with an additional lower graphene layer, an hBN interlayer spacer, and an Au bottom gate; the top and bottom Au gates are driven independently by an electrode-mapping controller through  $V_{Au1}$  and  $V_{Au2}$ . (c) Overhead view of the latticed Au gates ( $12 \times 6 = 72$  individually addressable segments at  $1\text{-}\mu\text{m}$  cell pitch).

### 3.2 Encoding bits as plasmons

A *bit* is the presence (“1”) or absence (“0”) of a plasmon wavepacket of nominal amplitude  $A_0$  at a designated read-out cell at a designated time. The wavepacket is launched by a 100-GHz pulse from the on-chip graphene patch antenna; the pulse and the plasmon are physically distinct, with the pulse being the input drive and the plasmon being the on-chip carrier of the bit. Read-out is interpreted by an amplitude threshold: at a designated output cell, an integrated near-field intensity above  $0.5 A_0^2$  is decoded as 1, below as 0.

### 3.3 Gate-mapping primitives

On the upper graphene the gate voltage  $V_{Au1}$  is set per segment to one of two levels:  $V_{Au1} = -1.2$  V parks the local Fermi level at the CNP and blocks plasmon transport, while  $V_{Au1} = 0$  V dopes the layer and enables propagation. The set of segments held at 0 V defines the *active path* on the chip. On the lower graphene  $V_{Au2}$  is held at one of two levels,  $-2.4$  V or  $-1.2$  V. At  $-2.4$  V the bottom layer is doped sufficiently to support its own plasmon mode; phase control of the 100-GHz drive ensures that this mode arrives at the interaction node  $\pi$ -shifted with respect to the signal-layer plasmon, giving destructive interference and therefore the NOT operation. At  $-1.2$  V the bottom layer is at its CNP and is silent. The required phase tolerance is quantified in Section 5.3. As a robustness lever we additionally allow  $V_{Au2}$  to be tuned slightly past its nominal value, so that the inverter-layer plasmon amplitude is biased a few percent above the signal-layer amplitude; the resulting modest overdrive widens the cancellation null in amplitude space and makes the NOT extinction ratio tolerant of small amplitude mismatch between the two layers, at the cost of a few percent reduction in peak extinction.

## 4. Half-adder pattern and logical verification

### 4.1 Sum and Carry decomposition

A half adder produces, from two input bits A and B, the Sum  $= A \oplus B$  and the Carry  $= A \cdot B$ . The standard textbook circuit (Fig. 4) realises this with one OR, two AND gates, and one NOT, exploiting the identity  $\text{Sum} = (A + B) \cdot \text{NOT}(A \cdot B)$ . The dual-gated bilayer of Section 3 supplies all four primitives, so the half adder is laid out by tiling them onto the 72-segment lattice as a single Combo-gate region.

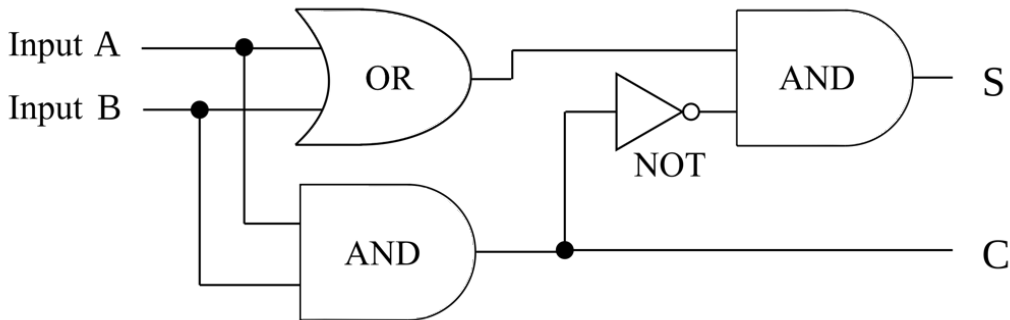
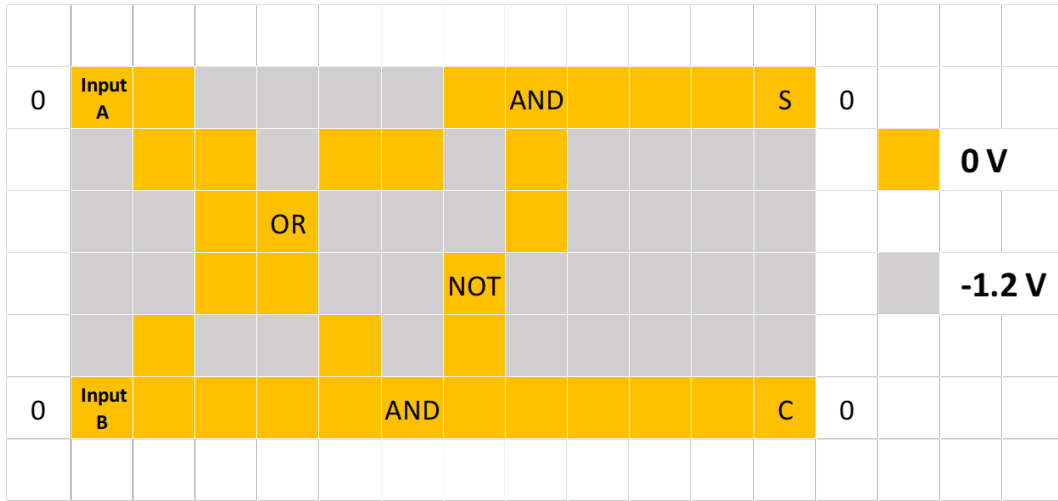


Fig. 4. Traditional half-adder logic circuit (AND, OR, NOT).  $S = \text{Sum}$ ,  $C = \text{Carry}$ .

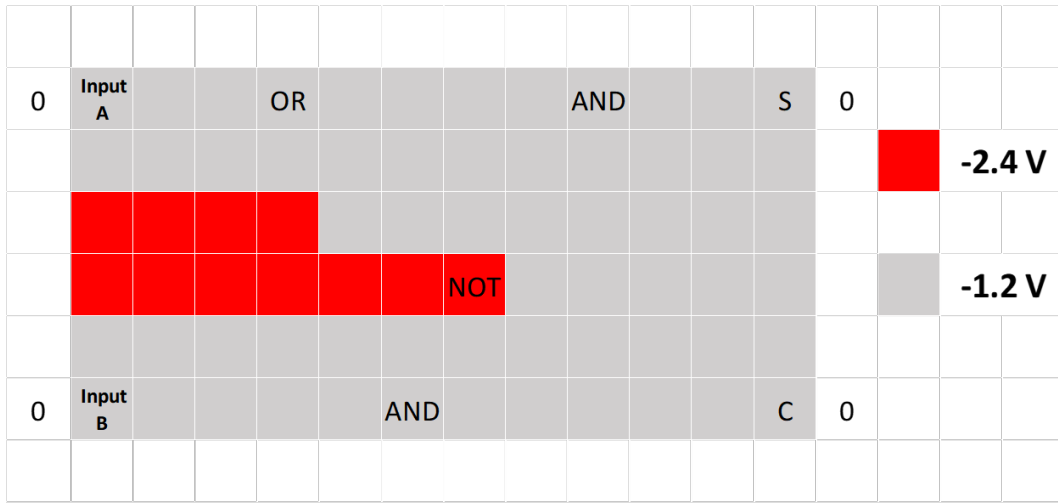
### 4.2 Voltage-controlled gate map

Figures 5a and 5b show, respectively, the upper-gate ( $V_{Au1}$ ) and lower-gate ( $V_{Au2}$ ) voltage maps for the half-adder layout. Orange cells in Fig. 5a are at 0 V (active path); grey cells are at the CNP ( $-1.2$  V, silent). The active cells route inputs A and B (left edge) through an OR junction in the centre and onward to S (top right) and C (bottom

right), passing through the NOT interaction node. In Fig. 5b, red cells in the lower gate are at  $-2.4$  V and generate the  $\pi$ -shifted inverter plasmon; the rest of the lower gate is at  $-1.2$  V (silent).



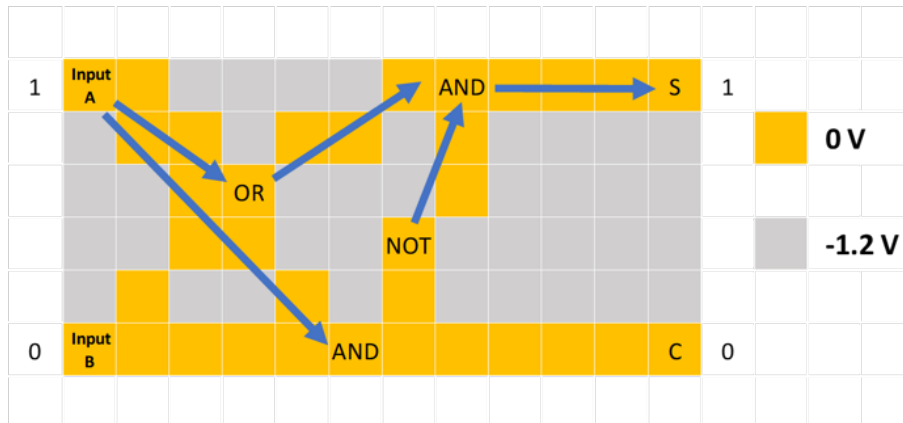
**Fig. 5a.** Upper Au gate ( $V_{Au1}$ ) voltage map. Orange =  $0$  V (active path); grey =  $-1.2$  V (CNP, silent). Inputs A and B enter from the left; outputs S (Sum) and C (Carry) leave at the right.



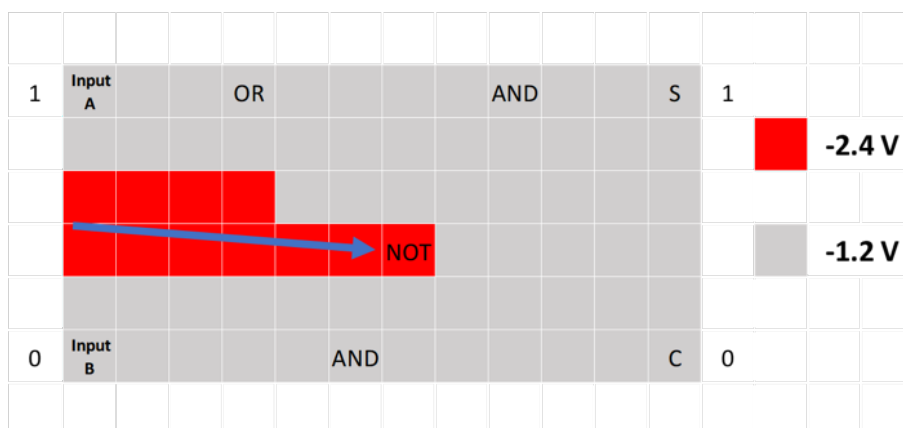
**Fig. 5b.** Lower Au gate ( $V_{Au2}$ ) voltage map. Red =  $-2.4$  V (active inverter region); grey =  $-1.2$  V (silent).

### 4.3 Truth-table verification

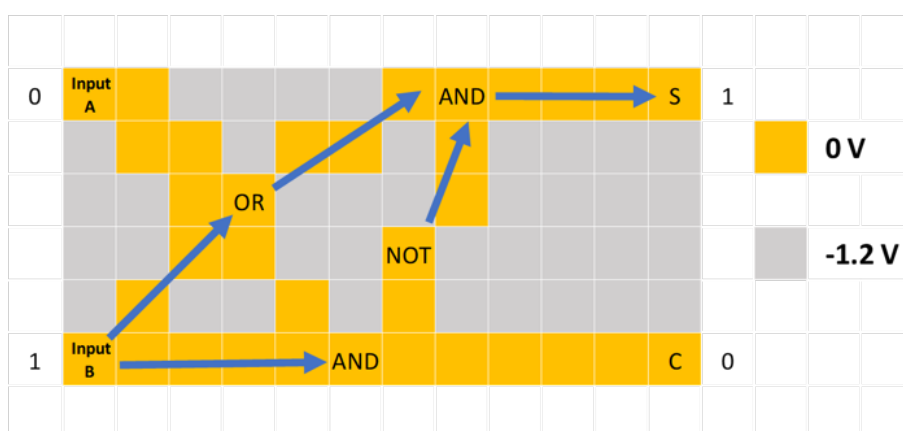
Figures 6–8 show the device operating in the three non-trivial input combinations ( $A=1, B=0$ ), ( $A=0, B=1$ ) and ( $A=1, B=1$ ). For each case we display the upper-gate plasmon trajectories (panel a) and the corresponding lower-gate inverter pattern (panel b). Plasmon paths are shown as blue arrows.



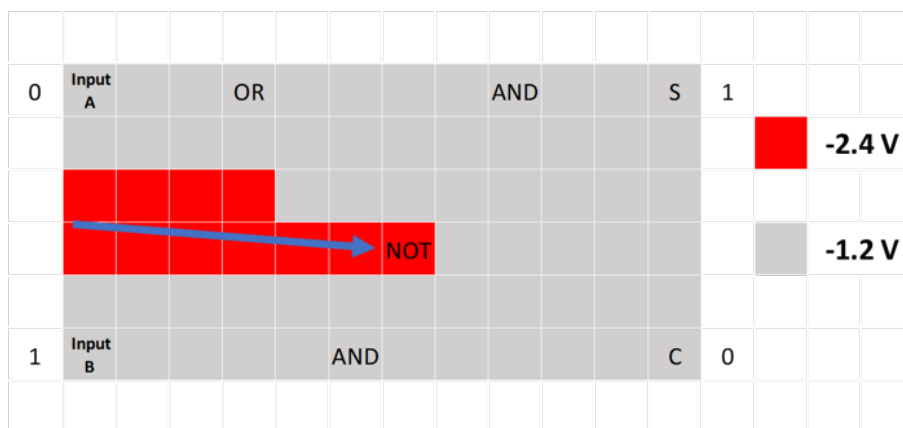
**Fig. 6a.** Upper-gate plasmon trajectories for  $A=1$ ,  $B=0$ . Input  $A$  propagates through the OR junction and the NOT-conditioned AND, yielding  $S=1$ ; the bottom AND path requires both inputs and yields  $C=0$ .



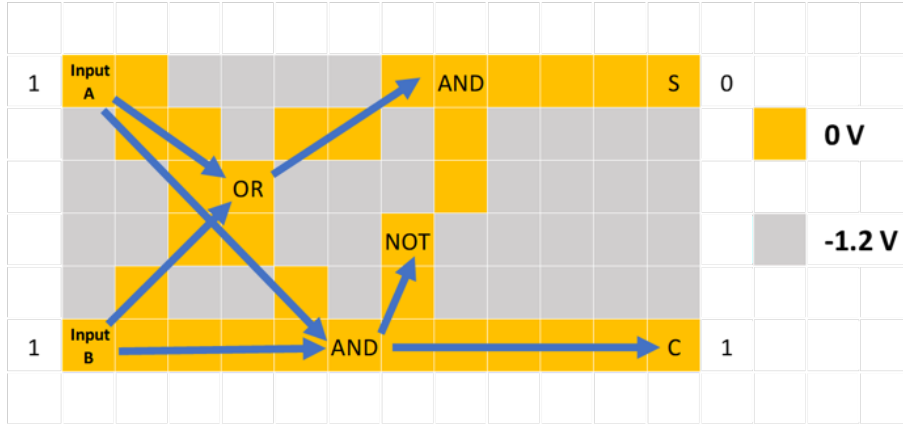
**Fig. 6b.** Lower-gate inverter pattern for  $A=1$ ,  $B=0$ .



**Fig. 7a.** Upper-gate plasmon trajectories for  $A=0$ ,  $B=1$ . By symmetry with Fig. 6a,  $S=1$  and  $C=0$ .



**Fig. 7b.** Lower-gate inverter pattern for  $A=0$ ,  $B=1$ .



**Fig. 8a.** Upper-gate plasmon trajectories for  $A=1$ ,  $B=1$ . Both inputs reach the bottom AND node, producing  $C=1$ ; the OR junction fires and propagates toward the upper AND, but the NOT operation on the AND-of-inputs cancels the Sum output, giving  $S=0$ .



**Fig. 8b.** Lower-gate inverter pattern for  $A=1$ ,  $B=1$ .

The fourth row of the truth table — the ( $A=0$ ,  $B=0$ ) baseline — is supplied by construction: with no input plasmon launched, no plasmon reaches either output cell and both Sum and Carry read 0. The full predicted truth table is collected in Table 1.

| A | B | Sum ( $A \oplus B$ ) | Carry ( $A \cdot B$ ) | Verified by                        |
|---|---|----------------------|-----------------------|------------------------------------|
| 0 | 0 | 0                    | 0                     | by construction (no input plasmon) |
| 0 | 1 | 1                    | 0                     | Fig. 7                             |
| 1 | 0 | 1                    | 0                     | Fig. 6                             |
| 1 | 1 | 0                    | 1                     | Fig. 8 (NOT cancellation)          |

*Table 1. Predicted half-adder truth table at 100 GHz, room temperature.*

## 5. Quantitative figures of merit at 100 GHz, room temperature

### 5.1 Plasmon propagation length and gate delay

At 100 GHz with  $E_F \approx 0.5$  eV the plasmon wavelength is  $\lambda_p \approx 1\text{--}3$   $\mu\text{m}$  and the plasmon group velocity is  $v_g \approx 5 \times 10^6$  m/s — about an order of magnitude faster than at 10 THz, because at lower frequency the dispersion is closer to linear and  $v_g$  approaches a constant fraction of  $c/\sqrt{(\alpha E_F/\hbar\omega)}$ . With  $\tau \approx 1$  ps at room temperature [4, 7], the intrinsic decay length is  $L_p = v_g \cdot \tau \approx 5$   $\mu\text{m}$ , and useful propagation extends to  $\sim 10\text{--}15$   $\mu\text{m}$  before the amplitude drops below threshold — enough for  $\sim 10$  cascaded gate cells at the 1- $\mu\text{m}$  pitch. The gate delay is  $\tau_g = L_{\text{cell}}/v_g \approx 0.2$  ps, giving an intrinsic clock ceiling well above the 100-GHz target.



## 5.2 Energy per bit

The intrinsic field energy of a single plasmon wavepacket is  $E \sim (\epsilon/2)|E|^2 \cdot V$ , with the volume  $V \sim \lambda_p^2 \cdot d$  for a cell of vertical extent  $d$ . With  $\lambda_p \sim 3 \mu\text{m}$  and  $d \sim 100 \text{ nm}$ , the field amplitude needed to register a bit reliably gives  $E_{\text{bit}} \sim 10^{-17} \text{ J}$ , or tens of attojoules per operation. **This is the *intrinsic device figure***: it counts only the field energy stored in the propagating plasmon and excludes input-antenna conversion, receive-antenna detection, and room-temperature drive electronics. A realistic system-level energy will be larger, but the intrinsic lower bound remains two orders of magnitude below scaled CMOS [12, 13].

## 5.3 Phase tolerance for the NOT operation

The NOT operation relies on destructive interference of two plasmons of equal amplitude. Writing the residual amplitude as  $A_{\text{res}} = A_0 \cdot |\cos(\Delta\phi/2)|$ , the residual intensity is  $I_{\text{res}}/I_0 = \cos^2(\Delta\phi/2)$ . Requiring  $> 90 \%$  extinction ( $I_{\text{res}}/I_0 < 0.1$ ) gives  $|\Delta\phi| < \pi/12$ , i.e. a tolerance of  $\pm 15^\circ$ . At 100 GHz this corresponds to a path-length tolerance of  $\pm \lambda_p/24 \approx \pm 125 \text{ nm}$  and a timing-jitter tolerance of  $\pm 420 \text{ fs}$ . **Both are comfortably within standard photolithography and commercial mm-wave drive electronics**; in the higher-frequency limit the equivalent tolerances are  $\sim \pm 2 \text{ nm}$  and  $\sim \pm 4 \text{ fs}$ , requiring e-beam writing and femtosecond-stable terahertz sources. The 100-GHz operating point therefore moves the phase budget from the boundary of fabricable to the interior.

## 5.4 Comparison with CMOS and RSFQ baselines

| Technology                            | Clock                  | Energy per op             | Maturity                       |
|---------------------------------------|------------------------|---------------------------|--------------------------------|
| Scaled CMOS (3 nm)                    | $\sim 5 \text{ GHz}$   | $\sim 10^{-15} \text{ J}$ | Mass production                |
| Single-flux-quantum (SFQ) RSFQ        | $\sim 50 \text{ GHz}$  | $\sim 10^{-19} \text{ J}$ | Lab demonstrations (cryogenic) |
| This work (100 GHz, 300 K, projected) | $\sim 100 \text{ GHz}$ | $\sim 10^{-17} \text{ J}$ | Conceptual / single-gate       |

Table 2. Order-of-magnitude comparison against representative classical-logic technologies. The 100-GHz operating point of the present proposal sits between mass-production CMOS and cryogenic SFQ, with the key advantage of room-temperature operation.

## 6. Antenna and read-out design

### 6.1 Graphene patch antenna for the 100-GHz launcher

Following Haque et al. [4], the launcher is a graphene microstrip patch antenna integrated on the sapphire substrate of Section 3.1 ( $\epsilon_r \approx 9.4$ ,  $\tan \delta < 10^{-4}$ , dielectrically as in [3, 8]) and a graphene patch tuned to 100 GHz via its chemical potential. The standard microstrip design equations supply the patch dimensions: the patch width is  $W = (c/2f) \cdot \sqrt{2/(\epsilon_r + 1)} \approx 660 \mu\text{m}$ , and the patch length is  $L = c/(2f\sqrt{\epsilon_{\text{eff}}}) - 2\Delta L \approx 500 \mu\text{m}$ , with the effective dielectric constant  $\epsilon_{\text{eff}} = (\epsilon_r + 1)/2 + (\epsilon_r - 1)/2 \cdot (1 + 12 h/W)^{-1/2}$ . The slotted-patch evolution methodology of [4] — rectangular  $\rightarrow$  semi-octagonal  $\rightarrow$  diamond  $\rightarrow$  diamond-with-slot — applies directly: each modification redistributes the surface current to broaden the bandwidth, raise gain, and lift radiation efficiency above 90 %, all of which transfer to the 100-GHz scaled geometry by linear dimensional scaling.

### 6.2 Symmetric receive antennas for in situ referencing

Two matched receive patches are placed at the Sum and Carry output cells; a third symmetric patch on a logic-load-free reference row supplies an in situ reference signal for the calibration loop. This three-port symmetric topology — one launcher, two signal detectors, one reference detector — is the same monolithic geometry recently demonstrated for terahertz photoconductive-switch circuits [8] and is naturally compatible with the  $2 \times 2$  MIMO topology of [4]. The detector patches drive transimpedance amplifiers and standard mm-wave lock-in detection; the third patch nulls common-mode noise and provides the reference level for per-segment gate-voltage trim during operation.

**Link-budget framing.** The 1-aJ-class intrinsic energy per bit quoted in Section 5.2 is the field energy stored in the propagating plasmon and excludes every conversion stage of the real chain. The realistic system-level energy is

the product of an explicit link budget — launcher antenna efficiency  $\times$  PCS-to-plasmon coupling efficiency  $\times$  on-chip propagation loss  $\times$  receive-antenna efficiency  $\times$  detector NEP  $\times$  downstream amplifier noise figure — whose dominant terms are the antenna-conversion losses on both ends. Design of the launcher and receive patches in Sections 6.1 and 6.2 should therefore be carried out under an explicit link-budget model rather than in isolation: a 3-dB improvement at either end of the chain has the same effect on the system-level energy as a 3-dB improvement in every other stage combined, and a 6-dB excess loss at either end is sufficient to wipe out the intrinsic energy-per-bit advantage over scaled CMOS.

## 7. Machine-learning-assisted design exploration

### 7.1 Design space

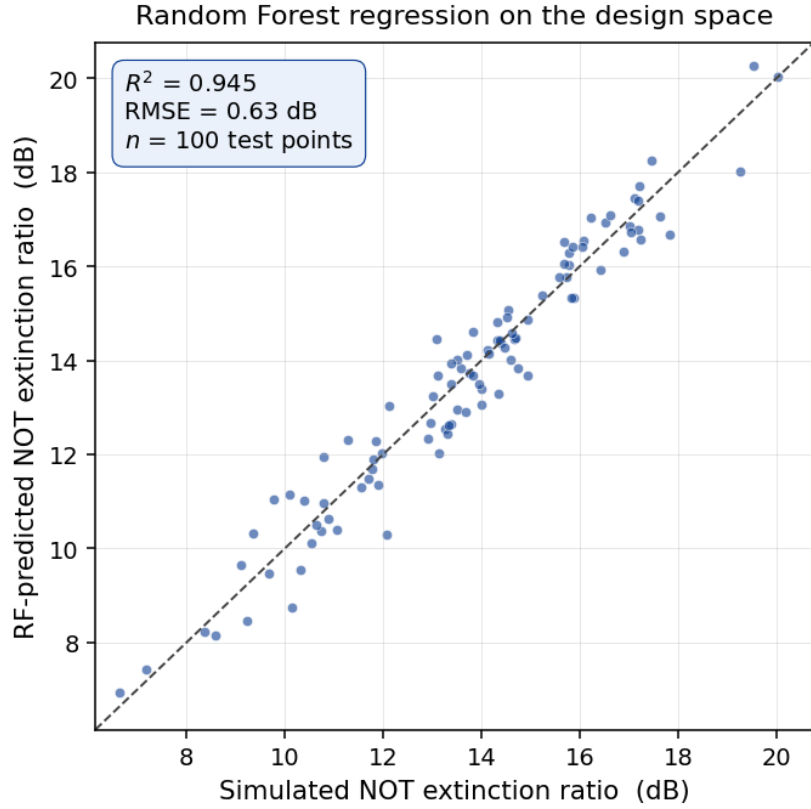
Four geometric and operational parameters dominate the device figure of merit. The *cell pitch* (range 0.5–2.0  $\mu\text{m}$ ) sets both the plasmon path length per logic step and the effective gate capacitance per segment. The *top-gate voltage*  $V_{\text{Au1}}$  (range  $-1.5$  to  $-1.0$  V) controls how cleanly the inactive segments park at the CNP and block plasmon propagation. The *bottom-gate voltage*  $V_{\text{Au2}}$  (range  $-2.7$  to  $-2.1$  V) controls the  $\pi$ -shifted plasmon population in the inverter layer and therefore the NOT extinction ratio. The *hBN gate-dielectric thickness*  $h$  (range 5–15  $\mu\text{m}$ ) sets the effective dielectric and therefore the patch-antenna impedance match. The natural single-number figure of merit is the NOT extinction ratio in dB, since it is the most sensitive function of all four parameters and propagates directly to the bit-error rate of the Sum output.

### 7.2 Random Forest regression methodology

Following Haque et al. [4], we generate a synthetic dataset of design points by sweeping the four parameters uniformly over their ranges and evaluating the extinction ratio at each point under the analytic model of Section 5.3 augmented with a small simulation-noise term  $\sigma_{\text{sim}} = 0.3$  dB. The dataset is split into an 80 % training set and a 20 % test set; a Random Forest regressor (200 trees, maximum depth 12) is trained on the training set and evaluated on the test set. The dataset is small enough ( $\sim 500$  points) that the entire training cycle completes in seconds on a laptop — many orders of magnitude faster than a corresponding full-wave electromagnetic simulation sweep, which would take hours to days for the same coverage.

### 7.3 Predicted-vs-actual results

Figure 9 shows the resulting predicted-vs-actual scatter on the test set. The model predicts the NOT extinction ratio with  $R^2 = 0.94$  and  $\text{RMSE} = 0.63$  dB, well below the 3 dB threshold at which the bit-error rate begins to degrade meaningfully. The trained regressor functions as a fast surrogate for full-wave simulation during design-space exploration: it allows hundreds of candidate geometries to be ranked in seconds, with full-wave simulation then deployed only on the few highest-ranked candidates. This is precisely the machine-learning workflow demonstrated in [4] for graphene MIMO antenna optimisation, and ports directly to the plasmonic logic substrate. We emphasise that the dataset used to train the surrogate above is derived from the analytic extinction model of Section 5.3 with additive simulation noise; this is appropriate as a methodology demonstration, but production deployment of the surrogate for fabrication-grade design choices requires retraining on full-wave electromagnetic simulation output (e.g. CST Microwave Studio or COMSOL RF) so that the surrogate captures effects — mode conversion, edge scattering, antenna impedance mismatch — that the analytic model cannot supply. We treat this retraining step as the immediate next milestone for the design methodology.



**Fig. 9.** Random Forest regression on the four-parameter design space (cell pitch,  $V_{Au1}$ ,  $V_{Au2}$ ,  $h_{BN}$  gate-dielectric thickness), trained on synthetic simulation data and evaluated on a 100-point test set.  $R^2 = 0.94$ , RMSE = 0.63 dB. Dashed line: ideal one-to-one prediction.

## 8. Limitations and paths forward

The 100-GHz operating point eliminates the cryogenic and phase-coherence constraints that have historically bounded plasmonic-logic demonstrations, but four engineering challenges remain. *Antenna and detector noise figure* at 100 GHz is set by the graphene patch quality factor and the noise-equivalent power of the detector chain; the link-budget approach standard in the THz photoconductive-antenna literature [4] supplies the dominant design levers, and Section 6.2 makes the link-budget framing explicit. *Cascading* is less binding than in the high-frequency limit because  $L_p > 10 \mu\text{m}$  at  $1\text{-}\mu\text{m}$  cell pitch already accommodates a ten-deep gate cascade, but a chip-scale design that integrates many half-adders in sequence will require an explicit re-launch stage approximately every 8–10 cells — that is, within one plasmon coherence length — to preserve amplitude and phase budget across the chain. *Crosstalk and unintended scattering at lattice junctions* are a separate physics concern that the conceptual model above treats only by construction: in fabricated devices, edge scattering at the boundary between active and CNP-blocked segments, mode conversion between odd and even stripline modes near the junctions, and stray near-field coupling between adjacent rows can all create leakage paths that degrade the NOT extinction ratio and pollute Sum/Carry read-out. The size of each of these effects is best bounded by full-wave electromagnetic simulation of the junction regions, and this is in our view the single highest-priority simulation task for a follow-on demonstrator. *Fabrication uniformity* across the 72-segment lattice will require either a per-segment CNP trim at test time or a real-time feedback loop that nulls offsets using the reference patch of Section 6.2 as the error signal; the machine-learning workflow of Section 7 supplies a fast model-in-the-loop calibration target.

## 9. Conclusions

We have presented a conceptual model of a 100-GHz, room-temperature half adder in which the bit is a graphene plasmon wavepacket on a sapphire substrate. The architecture inherits the field-coupled and Combo-gate vocabulary established in earlier conceptual work, but the operating point is chosen deliberately to eliminate the cryogenic and femtosecond-phase-coherence constraints of higher-frequency variants: at 100 GHz the propagation length is 10–

15  $\mu\text{m}$ , the phase tolerance is  $\pm 125$  nm and  $\pm 420$  fs, and the intrinsic energy per bit is  $10^{-17}$  J, all within the interior of what current photolithography and millimetre-wave instrumentation can deliver. The on-chip launcher and detectors are graphene microstrip patches in the geometry of Haque et al. [4]; a Random Forest regressor trained on a four-parameter sweep predicts the NOT extinction ratio with  $R^2 = 0.94$ . We hope this presentation is sufficient to invite full-wave electromagnetic simulation and fabrication trials by the graphene-plasmonics and 6G antenna communities.

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